

NITDGP/BTech/Reg/Even/2023-

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester End-term Examination, 2023-24

Course Code: CYC01

Course Name: Engineering Chemistry

Question Paper No.: NITD/CYC01/01

Full Marks: 60

Time: 180 Minutes

Instructions: 1. Answer all questions. 2. All the parts of a question must be written in same place. Otherwise, those answers will not be evaluated if written randomly.

Q. No.	Body of the Question	Marks	Mapped CO
1	(a) Draw a schematic diagram of Joule-Thomson experiment. Why Joule-Thomson experiment is called 'iso-enthalpic' process? What is 'inversion temperature'? (b) Calculate the entropy change when 10gm of Neon is heated from 27°C to 227°C at constant volume. ($C_v = 3 \text{ cal/mole}$). (c) Derive Clausius-Clayperon equation during the study of phase change of matter. Also explain why two ice cubes get fused when pressed together.	3+3+4=10	CO1
2	(a) For the reaction, $\text{H}_2 + \text{Br}_2 \longrightarrow 2\text{HBr}$, show that the order is <i>half</i> with respect to Br_2 at the initial stage of the reaction. (b) Under what condition 'steady state approximation' is valid? (c) "The oxidizing power of MnO_4^- varies with change of pH" Explain taking one example.	4+1+5=10	CO1
3	(a) For the complexes, $[\text{Co}(\text{H}_2\text{O})_6]^{+3}$, $[\text{Co}(\text{NH}_3)_6]^{+3}$ and $[\text{Co}(\text{CN})_6]^{-3}$, the magnitudes of Δ_o are found to be 20700, 22900 and 34800 cm^{-1} respectively. Account for the trend. (b) $\text{K}_2[\text{NiF}_6]$ is diamagnetic while $\text{K}_3[\text{CoF}_6]$ is paramagnetic. Both Ni(IV) and Co(III) are d^6 systems. How would you explain the difference in magnetic behaviour of these ions? (c) The d orbital splitting in octahedral coordination geometry is exactly opposite to the splitting in tetrahedral coordination geometry. Explain by showing the splitting in each case. (d) $\text{K}_2\text{Cr}_2\text{O}_7$ is dark orange in colour – Explain the fact.	3+2+3+2=10	CO4
4	(a) Calculate the number of valence electrons in the following complexes: $[\text{Ir}(\text{CO})\text{Cl}(\text{PPh}_3)_2]$; $[\text{Fe}_2(\text{CO})_9]$; $[(\eta^5\text{-C}_5\text{H}_5)_2\text{Ti}(\text{C}_2\text{H}_4)]$; (b) Draw the catalytic cycle for Monsanto acetic acid process mentioning all the steps involved in it. (c) Singlet oxygen is more toxic than the triplet form-Explain using molecular orbital theory.	3+4+3=10	CO4
5	(a) What is knocking of internal combustion engine? How tetraethyl lead (TEL) acts as an anti-knocking agent? What is the problem of using TEL as anti-knocking agent? How this problem can be reduced? (b) Calculate the wave number ($\tilde{\nu}$) of the C-H stretching from the following data; force constant, $k = 5 \times 10^5 \text{ g/s}^2$, mass of carbon atom = $20 \times 10^{-24} \text{ g}$, mass of hydrogen atom = $1.6 \times 10^{-24} \text{ g}$. (c) Write the names of various fractions (along with their uses) obtained during fractional distillation of crude oil.	5+2+3=10	CO3
6	(a) Write short notes on (i) PVC (ii) Teflon (iii) Bakelite. (b) Discuss the mechanism of polymerization reactions via free radical and cation intermediates with suitable example.	6+4=10	CO2